THE UNIVERSALITY OF WORDS $x^r y^s$ IN SYMMETRIC GROUPS

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ABSTRACT. We answer Problem 10.32 of the Kourovka Notebook, posed by Yu. I. Merzlyakov in 1986: for all nonzero integers r, s at least one of which is odd, the word $x^r y^s$ is universal on the symmetric group S_n — every permutation is a value — whenever $n \geq \max\{N_0, C \log m\}$, where $m = m(r, s)$ is the largest squarefree divisor of rs and C, N_0 are absolute constants. This is the symmetric-group analogue of the alternating-group theorem of Brenner, Evans and Silberger, and it replaces Droste's bound $n \geq \max\{6, 4m - 4\}$, linear in m and the best known since 1986. Even targets, and odd targets moving at least $n/2$ points, are factored as products of two cycles of explicitly controlled prime-related lengths, via the Herzog–Kaplan–Lev two-cycle criterion and Chebyshev-type prime counting; this part of the argument is explicit and fully effective. For odd targets moving fewer than $n/2$ points we show that, for adversarial r, s, any such two-cycle factorization would encode prime constellations of twin-prime type. These targets are instead factored, inside a suitable subset of $\{1, \ldots, n\}$, as a permutation of cycle type $(q_1, q_2, q_3, 1)$ times a full cycle of even length, via Boccara's criterion for products with a full cycle; the primes q_1, q_2, q_3, none dividing rs, come from Vinogradov's three-primes theorem relative to the excluded set of primes of rs — the one ineffective ingredient. A complementary construction respecting the parity hypothesis ($r = \operatorname{lcm}(1, \ldots, n)$, s its odd part) forces every value of the word to have all cycle lengths powers of two; the constant therefore cannot be taken to be 1, and the logarithmic shape is optimal for S_n, as it is for A_n.

1. Introduction

For which degrees n is every permutation in S_n a product of an r-th power and an s-th power? In the language of word maps: a word w in the free group on x, y is *universal* on a group G if the map $(x, y) \mapsto w(x, y)$ sends $G \times G$ onto G, and the question asks when the two-letter power word $x^r y^s$, $rs \neq 0$, is universal on S_n. Surjectivity questions for word maps are the noncommutative Waring problems of group theory: Ore's conjecture — every element of a finite non-abelian simple group is a commutator, that is, the word $[x, y]$ is universal — stood for nearly six decades before its resolution by Liebeck, O'Brien, Shalev and Tiep [13]. For an arbitrary nontrivial word w, the theorems of Larsen–Shalev and Larsen–Shalev–Tiep [11, 12] show that the values of w cover every sufficiently large such group in two steps: $w(G)^2 = G$. For the power words one can ask for more — genuine universality, in one step — and the classical literature on products of conjugacy classes of S_n makes the question concrete.

What no general theorem gives is uniformity in the word. For $w = x^r y^s$ the threshold $N(w)$ of [11] carries no dependence on (r, s) that one can read off, and the exponents can conspire against any fixed degree (§8 exhibits the conspiracy). The right question, posed forty years ago, is quantitative: how must the degree grow with the arithmetic of (r, s) for

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universality to set in? Write $m = m(r,s)$ for the largest squarefree divisor of rs (so m is the product of the distinct primes dividing rs), with the convention $m = 1$ when $r, s \in \{-1, 1\}$; the answer, it turns out, is governed by m alone, and is logarithmic in it.

Two papers in the same 1986 issue of the *Proceedings of the AMS* frame the problem. Brenner, Evans and Silberger [4] proved that for every constant $c > 8/5$ there is an $N_0(c)$ such that $x^r y^s$ is universal on A_n whenever $n \geq \max\{N_0(c),\, c \log m\}$; they showed moreover that the constant cannot be taken to be 1, so the logarithmic shape is optimal for A_n. Droste [5] proved, among other things, that when at least one of r, s is odd the word $x^r y^s$ is universal on S_n for $n \geq \max\{6,\, 4m - 4\}$ — a bound linear in m, hence exponentially weaker in $\log m$. Problem 10.32 of the Kourovka Notebook, contributed by Yu. I. Merzlyakov [10] to its tenth issue (1986) — the same year as the two *Proceedings* papers — asks:

> *Find an analogous* [*logarithmic*] *bound for the degree of the symmetric group S_n under hypothesis that at least one of the r, s is odd: up to now, here only a more crude bound* $n \geq \max\{6, 4m(r,s) - 4\}$ *is known* (*M. Droste, Proc. Amer. Math. Soc., 96, no. 1 (1986), 18–22*).

The parity hypothesis is necessary: if r and s are both even then every value $x^r y^s$ is an even permutation, and the word is not universal on S_n for any $n \geq 2$. This paper answers the problem in both directions: the logarithmic bound holds, and the logarithmic shape is optimal.

Main Theorem (answer to Kourovka Notebook Problem 10.32). There exist absolute constants C and N_0 with the following property. For all nonzero integers r, s at least one of which is odd, the word $x^r y^s$ is universal on the symmetric group S_n whenever

$$n \;\geq\; \max\{N_0,\; C \log m(r,s)\}.$$

The logarithmic threshold is the natural one. The word sees only the primes dividing rs: a cycle of length coprime to r is an r-th power (Lemma 2.2), so what the proofs must manufacture are cycle lengths up to n avoiding a prescribed set of primes. That set has total logarithmic mass $\log m = \sum_{p \mid rs} \log p$, while by Chebyshev the primes up to n have mass about n; so $n \asymp \log m$ is exactly the point at which the primes of rs can no longer swallow every prime below n. The lower bound of §8 shows the heuristic is sharp on the other side: taking $r = \operatorname{lcm}(1, \ldots, n)$ (and s its odd part) makes every prime up to n divide rs at cost $\log m = \theta(n)$, and $\theta(n) < n$ infinitely often — the arithmetic of rs really can cover every degree up to n.

The constant C produced by the proof is explicit but far from optimal; the threshold N_0 is ineffective, for the single reason that the proof invokes Vinogradov's three-primes theorem (see Remark 9.2). For even targets, and for odd targets moving at least $n/2$ points, the proof is fully explicit and effective and gives $n \geq \max\{N_0',\, 4.65 \log m\}$ with an explicit N_0'; it is only the remaining case — odd targets of small support — that requires analytic input. For the complementary lower bound, an adaptation of the Brenner–Evans–Silberger examples that respects the parity hypothesis (§8) shows the constant cannot be taken to be 1, so the logarithmic shape is optimal for S_n just as for A_n.

Why the symmetric group resisted. The argument of Brenner, Evans and Silberger runs through elements of order coprime to m: if every $z \in A_n$ is a product of two such elements, one wins by taking r-th and s-th roots inside the cyclic groups they generate. On S_n this mechanism breaks at once for *odd* targets: an odd permutation has at least one even-length cycle, so its order is even, and when $2 \mid m$ no odd permutation of order coprime to m exists at all. The two factors must therefore be treated asymmetrically — the first need only be an r-th power, the second an s-th power — and this is where the hypothesis enters: with s odd, even-length cycles of length coprime to s exist, and they carry the odd sign.

A second, subtler obstruction governs the proof: the first dictated the asymmetry of the factors; this one dictates their shape. Products of *two single cycles* are completely described by a criterion of Herzog, Kaplan and Lev (Theorem 2.3 below): z is a product of an l_1-cycle and an l_2-cycle essentially if and only if $l_1 + l_2$ is at least $\mathrm{m}(z) + \mathrm{c}(z)$ and of the same parity, and $|l_1 - l_2| \le \mathrm{m}(z) - \mathrm{c}(z)$, where $\mathrm{m}(z)$ is the number of moved points and $\mathrm{c}(z)$ the number of nontrivial cycles. For an odd target the two cycle lengths must have opposite parities, so $|l_1 - l_2|$ is odd and in particular nonzero; when $\delta(z) := \mathrm{m}(z) - \mathrm{c}(z)$ is small — think of a single transposition times a 3-cycle — the difference constraint forces the odd length u and even length v into an interval of bounded width, and u must avoid the prime divisors of r, v those of s. For adversarial r, s this demands, in effect, a pair of primes a bounded distance apart in prescribed positions — a constellation of twin-prime or binary-Goldbach strength, beyond reach. (Bounded gaps between primes are known after Zhang and Maynard, but not in prescribed windows, and the pair needed here involves the dilate $2^a q$ rather than a shift.) *No choice of two single cycles can be proved to work for all such targets short of solving such a constellation problem*; this is made precise in Proposition 6.1. The problem thus sits on the classical watershed of additive prime number theory: two cycle lengths give a binary constraint, of twin-prime strength, while a third summand would move the problem to ternary Goldbach, where Vinogradov's theorem waits.

Our escape buys that third summand with a different factor shape. For odd z of small support we write $z = AB$ inside an N-point subset, with $N = 2\tilde{q}$ for a prime $\tilde{q} \nmid s$. The factor B is a single N-cycle, hence an s-th power; the factor A, an r-th power, has cycle type $(q_1, q_2, q_3, 1)$ with q_1, q_2, q_3 primes summing to $N - 1$, each avoiding the primes of rs. The existence of such a factorization is governed by a beautiful criterion of Boccara (Theorem 2.4): a bound $\ell(\lambda) + \ell(\mu) \le N + 1$ — a genus bound, via the branched-cover correspondence noted in §2 — together with a parity condition. The genus bound demands exactly $\delta(z) \ge 3$ — and every odd non-transposition satisfies it — while the parity condition is automatic for odd z. The three primes are produced by Vinogradov's theorem *relative to an excluded set*: the number of representations $N - 1 = q_1 + q_2 + q_3$ is $\gg N^2/\log^3 N$, while a sieve bound, averaged over the $\asymp n/\log n$ admissible choices of $\tilde{q}$, shows that representations meeting the primes of rs are strictly fewer for at least one choice (indeed for most). The order of quantifiers is essential: r, s are fixed before $\tilde{q}$ is chosen, so the excluded set cannot correlate with N.

Three elementary observations make the case division clean. First, every nontrivial cycle moves at least two points, so $\mathrm{c}(z) \le \mathrm{m}(z)/2$ and hence $\delta(z) \ge \mathrm{m}(z)/2$: *small δ forces small support*, and the troublesome targets of the previous paragraph automatically leave at least half of $\{1, \dots, n\}$ free. Second, $\mathrm{sgn}(z) = (-1)^{\delta(z)}$, so δ is odd exactly for odd z, which is what makes the parity conditions in both criteria fall into place. Third, an odd z with $\delta(z) = 1$ is a bare transposition τ, and then $y = \tau$, $x = 1$ works outright since $\tau^s = \tau$ for odd s.

Structure of the proof. After reductions (§3) we may take s odd and both exponents at least 2 in absolute value. The theorem is then proved by three constructions:

(I) *z even, $z \ne 1$* (§4): $z = (q\text{-cycle}) \cdot (q\text{-cycle})$ for a single prime $q \nmid rs$ in $(3n/4, n]$. On the diagonal $l_1 = l_2$ the Herzog–Kaplan–Lev difference constraint is vacuous, and Chebyshev-type counting supplies the prime as soon as $n \ge 4.5 \log m$.

(II) *z odd,* $\mathrm{m}(z) \ge n/2$ (§5): $z = (u\text{-cycle}) \cdot (v\text{-cycle})$ with u a prime not dividing r in $(3n/4, n]$ and $v = 2^a q'$ for a prime $q' \nmid s$ in a dyadic rescaling $(3n/2^{a+2}, n/2^a]$ of the same window, $1 \le a \le 5$. Here $|u-v| < n/4 \le \delta(z)$ and $u+v > 3n/2 \ge \mathrm{m}(z) + \mathrm{c}(z)$, and the counting needs $n \ge 4.65 \log m$.

(III) *z odd,* $\mathrm{m}(z) < n/2$*,* $\delta(z) \ge 3$ (§6): the Boccara factorization described above, on an N-point set with $N = 2\tilde{q} \in (n/2, n]$.

Section 7 assembles the theorem, §8 proves the complementary lower bound, and §9 closes with a synthesis and remarks on the shapes of the factors, effectivity, and constants.

Relation to the literature. As discussed above, the general word-map theorems [11, 12] are qualitative in the parameters of the word: for $w = x^r y^s$ they give no uniformity of $N(w)$ in (r,s), and in particular no bound of the shape $n \gg \log m(r,s)$, which is the point of Problem 10.32. The complementary literature on explicit factorization — products of two prescribed conjugacy classes (Bertram [1]; Boccara [3]; Herzog, Kaplan and Lev [9]; the survey material in [15]) — provides exactly the combinatorial control needed here, and this paper joins those criteria to classical analytic number theory at a single point. The content lies in locating that point — Proposition 6.1 shows the analytic input is unavoidable for any two-single-cycle approach, not merely convenient — and in choosing a factor shape for which the input needed is ternary rather than binary.

2. Preliminaries

2.1. **Notation.** Throughout, n denotes the degree of the symmetric group S_n and $z \in S_n$ a target permutation. We write $\mathrm{supp}(z)$ for the set of moved points, $\mathrm{m}(z) = |\mathrm{supp}(z)|$, $\mathrm{c}(z)$ for the number of cycles of length at least 2, and

$$\delta(z) \;=\; \mathrm{m}(z) - \mathrm{c}(z).$$

A *cycle type* on an N-point set Ω is a partition $\lambda \vdash N$ in which fixed points are recorded as parts equal to 1; we write $\ell(\lambda)$ for the total number of parts, fixed points included, and C_λ for the conjugacy class of $\mathrm{Sym}(\Omega)$ with that type. For r, s we set $m = m(r,s)$ as in the introduction and

$$L \;=\; \log m \;=\; \sum_{p \mid rs} \log p,$$

sums and products over p always ranging over primes. All logarithms are natural.

Lemma 2.1. *For every $z \in S_n$:*

1. $\mathrm{sgn}(z) = (-1)^{\delta(z)}$*; in particular z is odd if and only if $\delta(z)$ is odd.*
2. $\mathrm{c}(z) \le \mathrm{m}(z)/2$*, hence* $\delta(z) \ge \mathrm{m}(z)/2 \ge \mathrm{c}(z)$*.*
3. *If z is odd and $\delta(z) = 1$ then z is a transposition.*

Proof. An ℓ-cycle has sign $(-1)^{\ell-1}$; multiplying over the nontrivial cycles of z, whose lengths sum to $\mathrm{m}(z)$ and whose number is $\mathrm{c}(z)$, gives (1). Each nontrivial cycle moves at least 2 points, which is (2). For (3), $\delta(z) = 1$ and (2) give $\mathrm{m}(z) \le 2$, so z moves exactly two points. $\square$

2.2. **Roots.** The next lemma is the standard fact about powers of cycles (see, e.g., [21, §4.8]: a cycle of length ℓ raised to the t-th power is a product of $\gcd(\ell, t)$ cycles of length $\ell/\gcd(\ell,t)$).

Lemma 2.2. *Let $\sigma \in S_n$ be a product of pairwise disjoint cycles of lengths $\ell_1, \ldots, \ell_k$, and let t be a nonzero integer with* $\gcd(\ell_i, t) = 1$ *for all i. Then there is $x \in S_n$ with* $\mathrm{supp}(x) = \mathrm{supp}(\sigma)$ *and $x^t = \sigma$.*

Proof. It suffices to treat a single ℓ-cycle γ and multiply the resulting roots, which are disjoint. Choose a with $at \equiv 1 \pmod{\ell}$; then $(\gamma^a)^t = \gamma^{at} = \gamma$ since $\gamma^\ell = 1$, and γ^a is again an ℓ-cycle on the same support because $\gcd(a, \ell) = 1$. □

2.3. **Two factorization criteria.** The first criterion characterizes products of two single cycles. It was proved by Herzog, Kaplan and Lev; the case of two cycles of given lengths goes back to Boccara [2], and the two-l-cycle case to Bertram [1].

Theorem 2.3 (Herzog–Kaplan–Lev [9, Theorem 7])**.** *Let $\sigma \in S_n$, $\sigma \neq 1$, and let $n \geq l_1 \geq l_2 \geq 2$ be integers. Then σ is a product of an l_1-cycle and an l_2-cycle in S_n if and only if one of the following holds:*

(a) *σ has exactly two nontrivial cycles, and their lengths are l_1 and l_2; or*
(b) $l_1 + l_2 \geq \mathrm{m}(\sigma) + \mathrm{c}(\sigma)$, $\quad l_1 + l_2 \equiv \mathrm{m}(\sigma) + \mathrm{c}(\sigma) \pmod 2$, $\quad$ *and* $\quad l_1 - l_2 \leq \mathrm{m}(\sigma) - \mathrm{c}(\sigma)$.

We use the sufficiency of (b) throughout; the necessity direction enters only in the obstruction Proposition 6.1. Note that if $\sigma = AB$ then also $\sigma = B\,(B^{-1}AB)$, so the two lengths may be realized in either order; we use this freedom silently to put the r-th-power factor first.

The second criterion characterizes products of an arbitrary class with the class of full cycles. It is due to Boccara [3]; it is also equivalent, via the Hurwitz monodromy correspondence, to the realizability of branched covers of the sphere with three branch points one of which is totally ramified, proved by Thom for the sphere and by Edmonds, Kulkarni and Stong in general [18, 6] (see [15, Theorem 2.1] for a modern statement).

Theorem 2.4 (Boccara [3])**.** *Let Ω be a set of size $N \geq 2$, let $\lambda \vdash N$ be a cycle type (fixed points counted as parts 1), and let $z \in \mathrm{Sym}(\Omega)$ have cycle type $\mu \vdash N$. Then $z = AB$ for some $A \in C_\lambda$ and some N-cycle B on Ω if and only if*

$$\ell(\lambda) + \ell(\mu) \;\leq\; N + 1 \qquad \textit{and} \qquad \ell(\lambda) + \ell(\mu) \;\equiv\; N + 1 \pmod 2.$$

Remark 2.5. Both criteria have been verified mechanically by exhaustive computation: Theorem 2.3 over all of S_7 and S_8 (all 1618 triples (cycle type, l_1, l_2) with $2 \leq l_1, l_2 \leq n$, the identity type included as a sanity case), and Theorem 2.4 for all $N \leq 8$ (all ordered pairs (λ, μ) of types, both directions of the equivalence), with no discrepancies. We mention this because both statements are used here in load-bearing ways and details of [3] circulate imprecisely — its volume number, for instance, appears incorrectly in more than one recent bibliography.

2.4. **Prime counting.** Let $\theta(x) = \sum_{p \leq x} \log p$ and $\pi(x)$ the usual counting functions. We use the classical explicit bounds of Rosser and Schoenfeld: (2.1) is [16, Theorem 9, eq. (3.32)], (2.3) and (2.4) are [16, eqs. (3.6) and (3.5)] respectively, and (2.2) is from their later paper [17, Corollary to Theorem 6]:

$$\theta(x) \;<\; 1.01624\,x \qquad \text{for all } x > 0, \tag{2.1}$$

$$\theta(x) \;>\; 0.985\,x \qquad \text{for } x > 11927, \tag{2.2}$$

$$\pi(x) \;<\; 1.25506\,\frac{x}{\log x} \qquad \text{for } x > 1, \tag{2.3}$$

$$\pi(x) \;>\; \frac{x}{\log x} \qquad \text{for } x \geq 17. \tag{2.4}$$

Lemma 2.6 (Prime windows)**.** *Let $0 < \alpha < \beta \leq 1$ and put $K = 0.985\,\beta - 1.01624\,\alpha$, assumed positive. Let Q be any set of primes with $\sum_{q \in Q} \log q \leq L$. If $n > 11927/\beta$ and $Kn > L$, then the interval $(\alpha n,\, \beta n]$ contains a prime not belonging to Q.*

Proof. By (2.1) and (2.2), $\sum_{\alpha n<p\le\beta n}\log p=\theta(\beta n)-\theta(\alpha n)>0.985\beta n-1.01624\alpha n=Kn>L$. The primes of Q in the window contribute at most $\sum_{q\in Q}\log q\le L$ to the left side, so some prime of the window lies outside Q. □

We record the windows we use, with K rounded down:

$$\begin{gathered}(\tfrac{3}{4}n,\, n]:\;\; K=0.2228;\\ \text{rescaled, } (3n/2^{a+2},\, n/2^a]:\;\; K=0.2228/2^a \quad (a\ge 1,\; n>11927\cdot 2^a).\end{gathered}\tag{2.5}$$

In particular the base window contains a prime avoiding any prescribed set of primes of log-weight at most L provided $n\ge 23856$ and $n\ge 4.5\,L$; the rescaled windows are used, five at a time, in §5.

Lemma 2.7. *For every integer $m\ge 2$ with $L=\log m$,*

$$\omega(m)\;\le\;1.28\,\frac{L}{\log L}\;+\;1429,$$

where $\omega(m)$ is the number of distinct prime divisors of m (and the first term is read as 0 when $L\le 1$).

Proof. Let $\omega=\omega(m)$ and let q_ω denote the ω-th prime. The i-th smallest prime divisor of m is at least the i-th prime, so $L\ge\theta(q_\omega)$. If $q_\omega\le 11927$ then $\omega=\pi(q_\omega)\le\pi(11927)=1429$. Otherwise $q_\omega>11927$ and (2.2) gives $q_\omega\le L/0.985\le 1.0153\,L$, and (2.3) gives $\omega=\pi(q_\omega)\le\pi(1.0153L)<1.25506\cdot 1.0153\,L/\log(1.0153L)<1.28\,L/\log L$. □

2.5. **Analytic input.** For a positive integer k let

$$\begin{aligned}r_2(k)&=\#\{(p_1,p_2):\; p_i \text{ prime},\; p_1+p_2=k\},\\ r_3(k)&=\#\{(p_1,p_2,p_3):\; p_i \text{ prime},\; p_1+p_2+p_3=k\}\end{aligned}$$

count ordered representations as sums of primes. We use the following two classical facts.

Theorem 2.8 (Vinogradov [20]; see [19, Theorem 3.4])**.** *There are constants $c_1>0$ and M_0 such that $r_3(M)\ge c_1M^2/\log^3 M$ for every odd $M\ge M_0$.*

Theorem 2.9 (Sieve bound; see [7, Theorem 3.11])**.** *There is an absolute constant C_2 such that for every even $k\ge 4$,*

$$r_2(k)\;\le\;C_2\,S(k)\,\frac{k}{\log^2 k},\qquad \textit{where}\quad S(k)\;=\;\prod_{p\mid k,\;p>2}\frac{p-1}{p-2}.$$

For odd k, trivially $r_2(k)\le 2$ (one summand must be 2).

We also use the Brun–Titchmarsh inequality in the form of Montgomery and Vaughan [14, Theorem 2]: for coprime a,d and $x>d$,

$$\pi(x;d,a)\;<\;\frac{2x}{\varphi(d)\log(x/d)},\tag{2.6}$$

where $\pi(x;d,a)$ counts the primes $p\le x$ with $p\equiv a \pmod d$.

Remark 2.10. Theorem 2.8 is the one ineffective ingredient of this paper: M_0 is not computable from the classical proof (Siegel–Walfisz). The constant c_1 is an explicit number, and effective versions of the three-primes theorem exist (Helfgott [8]); see Remark 9.2.

3. Reductions

Lemma 3.1. *Let G be a group and r, s nonzero integers.*

(1) *x^ry^s is universal on G if and only if x^sy^r is.*
(2) *Universality of x^ry^s on G depends only on $(|r|, |s|)$.*
(3) *If $|r| = 1$ or $|s| = 1$ then x^ry^s is universal on every group.*

Proof. (1) Given $z \in G$, apply universality of x^sy^r to z^{-1}: if $z^{-1} = a^sb^r$ then $z = b^{-r}a^{-s} = (b^{-1})^r(a^{-1})^s$. (2) Substitute $x \mapsto x^{-1}$, $y \mapsto y^{-1}$ as needed. (3) If $|s| = 1$ take $x = 1$, $y = z^{\pm 1}$; if $|r| = 1$ take $y = 1$, $x = z^{\pm 1}$. □

Since $m(r, s) = m(s, r)$ and m depends only on $(|r|, |s|)$, Lemma 3.1 allows us to assume for the remainder of the paper:

$$r \geq 2, \qquad s \geq 3 \text{ odd}, \tag{3.1}$$

the excluded cases being trivially universal. (If the odd exponent among r, s is the first one, swap by Lemma 3.1(1); then drop signs by (2); if either exponent is 1 use (3), which in particular covers $m = 1$.)

Two families of targets are handled at once. If $z = 1$, take $x = y = 1$. If z is a transposition τ, take $x = 1$ and $y = \tau$: since s is odd, $y^s = \tau^s = \tau = z$. By Lemma 2.1(3) these dispose of all odd z with $\delta(z) = 1$.

4. Even targets

Proposition 4.1. *Assume* (3.1)*, $n \geq 23856$ and $n \geq 4.5\,L$. Then every even $z \in S_n$, $z \neq 1$, satisfies $z = x^ry^s$ for some $x, y \in S_n$.*

Proof. Under (3.1) the odd number $s \geq 3$ contributes an odd prime to m, so $m \geq 3$, $L \geq \log 3 > 0$, and $Kn \geq 0.2228 \cdot 4.5\,L > L$ for the window $(\frac{3}{4}n, n]$ of (2.5). By Lemma 2.6, applied to this window with Q the set of primes dividing rs, there is a prime

$$q \in (\tfrac{3}{4}n,\, n], \qquad q \nmid rs.$$

We check condition (b) of Theorem 2.3 for $l_1 = l_2 = q$. Since $\mathrm{c}(z) \leq \mathrm{m}(z)/2$ (Lemma 2.1) and $\mathrm{m}(z) \leq n$,

$$l_1 + l_2 = 2q > \tfrac{3}{2}n \;\geq\; \mathrm{m}(z) + \mathrm{c}(z).$$

As z is even, $\delta(z) = \mathrm{m}(z) - \mathrm{c}(z)$ is even, so $\mathrm{m}(z) + \mathrm{c}(z)$ is even as well, matching $l_1 + l_2 = 2q$. Finally $l_1 - l_2 = 0 \leq \mathrm{m}(z) - \mathrm{c}(z)$, the latter being positive since $z \neq 1$. Hence $z = AB$ with A, B both q-cycles. Since $q \nmid rs$, Lemma 2.2 provides x with $x^r = A$ and y with $y^s = B$, and $x^ry^s = AB = z$. □

5. Odd targets of large support

Proposition 5.1. *Assume* (3.1)*, $n \geq 381700$ and $n \geq 4.65\,L$. Then every odd $z \in S_n$ with $\mathrm{m}(z) \geq n/2$ satisfies $z = x^ry^s$ for some $x, y \in S_n$.*

Proof. Write $\delta = \mathrm{m}(z) - \mathrm{c}(z)$. By Lemma 2.1(2), $\delta \geq \mathrm{m}(z)/2 \geq n/4$. As in Proposition 4.1, $L > 0$, so $0.2228\,n \geq 0.2228 \cdot 4.65\,L > L$, and Lemma 2.6 applied to the window $(\frac{3}{4}n, n]$ with $Q = \{p : p \mid r\}$ yields a prime

$$u \in (\tfrac{3}{4}n,\, n], \qquad u \nmid r$$

(note u is odd since $u > 3$).

For the even factor we use the dyadic windows of (2.5). Suppose that for every $a \in \{1, \ldots, 5\}$, every prime in $W_a = (3n/2^{a+2},\, n/2^a]$ divides s. The intervals W_a are pairwise disjoint (the top of W_{a+1} is $n/2^{a+1} < 3n/2^{a+2}$, the bottom of W_a), and $n/2^a > 11927$ for $a \le 5$ since $n \ge 381700 > 11927 \cdot 2^5$; so, applying (2.1) and (2.2) to each window and summing,

$$L \;\ge\; \sum_{a=1}^{5}\bigl(\theta(n/2^a) - \theta(3n/2^{a+2})\bigr) \;\ge\; \sum_{a=1}^{5} \frac{0.2228}{2^a}\, n \;=\; \frac{31}{32} \cdot 0.2228\, n \;>\; 0.2158\, n,$$

contradicting $n \ge 4.65\, L$ (as $4.65 \cdot 0.2158 > 1$). Hence there are $a \le 5$ and a prime $q' \in W_a$ with $q' \nmid s$; put

$$v \;=\; 2^a q' \;\in\; (\tfrac{3}{4}n,\, n],$$

an even number with $\gcd(v, s) = 1$ (s is odd and $q' \nmid s$).

We check Theorem 2.3(b) for the pair $\{u, v\}$ (in whichever order makes the lengths nonincreasing). First, $u + v > \frac{3}{2}n \ge \frac{3}{2}\mathrm{m}(z) \ge \mathrm{m}(z) + \mathrm{c}(z)$, so $u + v > \mathrm{m}(z) + \mathrm{c}(z)$. Second, $u + v$ is odd (one length even, one odd) and $\mathrm{m}(z) + \mathrm{c}(z) \equiv \mathrm{m}(z) - \mathrm{c}(z) = \delta \pmod 2$ is odd as well since z is odd; so $u + v \ge \mathrm{m}(z) + \mathrm{c}(z)$ with the correct parity. Third, since $u, v \in (\frac{3}{4}n,\, n]$,

$$|u - v| \;<\; \tfrac{1}{4}n \;\le\; \delta \;=\; \mathrm{m}(z) - \mathrm{c}(z).$$

Both lengths are at least 2 and at most n, and $z \ne 1$ as z is odd. Hence $z = AB$ with A a u-cycle and B a v-cycle. Since $\gcd(u, r) = 1$ and $\gcd(v, s) = 1$, Lemma 2.2 gives $x^r = A$, $y^s = B$, so $z = x^r y^s$. □

6. Odd targets of small support: obstruction and escape

This case is the heart of the paper: it is here that the two-single-cycle method provably demands twin-prime-type input (Proposition 6.1 below), and here that the ternary construction enters. Throughout this section we assume (3.1) and consider odd targets z with

$$\mathrm{m}(z) < n/2, \qquad \delta(z) \ge 3.$$

6.1. **The two-cycle obstruction.** We first make precise the claim of the introduction that no analogue of Propositions 4.1 and 5.1 — two single cycles, one an r-th power and one an s-th power — can cover the present case. This is the one place where the necessity direction of Theorem 2.3 is used; nothing later depends on it.

Proposition 6.1 (two-cycle obstruction)**.** *Let r be even and s odd, and suppose every prime $p \le x$ divides r and every odd prime $p \le x$ divides s, where $x \ge 5$. Let $z \in S_n$ be a 4-cycle, and suppose $z = AB$ where A is an l_1-cycle that is an r-th power in S_n and B is an l_2-cycle that is an s-th power ($l_1, l_2 \ge 2$). Then l_1 is odd with every prime factor exceeding x, l_2 is even with every odd prime factor exceeding x, and $|l_1 - l_2| \le 3$. In particular, if $n < x^2$, then l_1 is a prime and l_2 is 2^a or $2^a q$ with $q > x$ prime.*

Proof. An l-cycle is a t-th power in S_n if and only if $\gcd(l, t) = 1$: each cycle of w of length ℓ contributes $\gcd(\ell, t)$ cycles of length $\ell/\gcd(\ell, t)$ to w^t, so a single l-cycle arises only from a single ℓ-cycle with $\ell = l$ and $\gcd(l, t) = 1$. Hence $\gcd(l_1, r) = 1$ and $\gcd(l_2, s) = 1$, giving the prime-factor claims. Since r is even, A is an even permutation, so l_1 is odd; since z is odd, l_2 is even. A 4-cycle has one nontrivial cycle, so case (a) of Theorem 2.3 is impossible, and the necessity of case (b) gives $|l_1 - l_2| \le \mathrm{m}(z) - \mathrm{c}(z) = 3$. If $n < x^2$, a length at most n whose (odd) prime factors all exceed x has at most one of them. □

Taking $r = \prod_{p \le x} p$ and $s = \prod_{2 < p \le x} p$, so that $\log m = \theta(x) \asymp x$, a two-single-cycle proof of the Main Theorem would thus produce, for every sufficiently large x, primes $u, q \in (x, n]$ with $u - 2^a q \in \{\pm 1, \pm 3\}$ (or u within 3 of a power of 2), where $n \asymp x$: a constellation problem of twin-prime type, involving the dilate $2^a q$ rather than a shift, and beyond any current method. (Of course the 4-cycle itself is an s-th power for odd s — the obstruction concerns the two-single-cycle method with both cycles nontrivial, not membership in the image of the word map.)

6.2. **The construction, given a good prime.** Call a prime $\tilde{q}$ *good* (for the fixed data r, s, n) if

(G1) $n/4 < \tilde{q} \le n/2$ and $\tilde{q} \nmid s$;
(G2) the odd number $M = 2\tilde{q} - 1$ admits a representation $M = q_1 + q_2 + q_3$ with q_1, q_2, q_3 primes none of which divides rs.

Lemma 6.2. *If a good prime $\tilde{q}$ exists, then every odd $z \in S_n$ with $\mathrm{m}(z) < n/2$ and $\delta(z) \ge 3$ satisfies $z = x^r y^s$.*

Proof. Put $N = 2\tilde{q} \in (n/2, n]$; since $\mathrm{m}(z) < n/2 < N \le n$, we may choose $\Omega \subseteq \{1, \dots, n\}$ with $\mathrm{supp}(z) \subseteq \Omega$ and $|\Omega| = N$. Let $q_1 + q_2 + q_3 = N - 1$ be as in (G2) and set

$$\lambda = (q_1, q_2, q_3, 1) \ \vdash \ N, \qquad \ell(\lambda) = 4.$$

Let $\mu \vdash N$ be the type of z on Ω; it consists of the $\mathrm{c}(z)$ nontrivial cycles of z together with $N - \mathrm{m}(z)$ fixed points, so $\ell(\mu) = \mathrm{c}(z) + N - \mathrm{m}(z) = N - \delta(z)$. Then

$$\ell(\lambda) + \ell(\mu) \ = \ N + 4 - \delta(z) \ \le \ N + 1 \iff \delta(z) \ge 3,$$

which holds; and modulo 2,

$$\ell(\lambda) + \ell(\mu) \ = \ N + 4 - \delta(z) \ \equiv \ N + 1 \iff \delta(z) \equiv 3 \equiv 1,$$

which holds since z is odd (Lemma 2.1(1)). By Theorem 2.4 there are $A \in C_\lambda$ and an N-cycle B on Ω with $AB = z$ (using, as noted after Theorem 2.3, the freedom to put the C_λ-factor first). The nontrivial cycles of A have prime lengths q_1, q_2, q_3, none dividing r, so Lemma 2.2 gives x with $x^r = A$. The factor B is an N-cycle with $N = 2\tilde{q}$, and $\gcd(N, s) = 1$ since s is odd and $\tilde{q} \nmid s$; Lemma 2.2 gives y with $y^s = B$. Extending by the identity off Ω, $x^r y^s = AB = z$. □

6.3. **Averaging the singular series.** It remains to produce a good prime. For any *single* candidate M the sieve bound of Theorem 2.9 loses a factor $\asymp \log n$ against $r_3(M)$ when rs has many prime divisors (see Remark 6.6); the remedy is to average over a full window of candidates. Let

$$W \ = \ \{\tilde{q} \text{ prime}: \ (n+2)/4 < \tilde{q} \le n/2\},$$

the lower endpoint chosen so that $M = 2\tilde{q} - 1 > n/2$ for every $\tilde{q} \in W$.

Lemma 6.3. *For $n \ge 10^8$, $|W| \ \ge \ 0.15\, n/\log n$.*

Proof. By (2.4), $\pi(n/2) > (n/2)/\log(n/2) > n/(2\log n)$. By (2.3) and $\log(n/4) \ge 0.9 \log n$ (valid for $n \ge 4^{10}$), $\pi((n+2)/4) \le \pi(n/4) + 1 < 1.25506\,(n/4)/(0.9\log n) + 1 < 0.349\, n/\log n$. Subtract. □

Lemma 6.4 (Singular series on average)**.** *There is an absolute constant C_3 (one may take $C_3 = 4$ for $n \ge 10^8$) such that for every integer b with $0 \le b \le n+2$,*

$$\sum_{\substack{\tilde{q} \in W \\ 2\tilde{q} - b \text{ even}, \ \ge 4}} S(2\tilde{q} - b) \ \le \ C_3 \frac{n}{\log n},$$

with $S(\cdot)$ as in Theorem 2.9.

Proof. For even $k \geq 4$ expand the product: writing g for the multiplicative function supported on odd squarefree integers with $g(p) = 1/(p-2)$,

$$S(k) \;=\; \prod_{p|k,\, p>2}\Big(1+\frac{1}{p-2}\Big) \;=\; \sum_{\substack{d|k\\ d\text{ odd squarefree}}} g(d).$$

Write Σ for the sum in the statement; its arguments $k = 2\tilde{q} - b$ all satisfy $4 \leq k \leq n$, so every divisor of such a k is at most n. Hence, interchanging sums,

$$\Sigma \;\leq\; \sum_{\substack{d\leq n\\ d\text{ odd squarefree}}} g(d)\,\#\{\tilde{q} \in W:\; 2\tilde{q} \equiv b \pmod d\}.$$

For each odd d the congruence $2\tilde{q} \equiv b \pmod d$ determines a single residue class a_d modulo d. Split at $D = n^{1/3}$.

If $d \leq D$: when $\gcd(a_d, d) > 1$, at most $\omega(d) \leq \log d$ primes lie in the class; when $\gcd(a_d, d) = 1$, Brun–Titchmarsh (2.6) with $x = n/2$ gives, for $n \geq 10^8$,

$$\#\{\cdot\} \;\leq\; \pi(n/2; d, a_d) \;<\; \frac{n}{\varphi(d)\log(n/(2d))} \;\leq\; \frac{n}{0.6\,\varphi(d)\log n}$$

using $\log(n/(2d)) \geq \frac{2}{3}\log n - \log 2 \geq 0.6\log n$. Since $\sum_{d\text{ odd squarefree}} g(d)/\varphi(d) = \prod_{p>2}\big(1+\frac{1}{(p-2)(p-1)}\big) < 1.75$, the total contribution of $d \leq D$ is at most $\frac{1.75}{0.6}\,n/\log n + \sum_{d\leq D} g(d)\log d \leq 2.92\,n/\log n + e^{1.2}\log^2 D$, using $\sum_{d\leq x} g(d) \leq \prod_{p\leq x,\,p>2}(1+\frac{1}{p-2}) \leq e^{1.2}\log x$ for $x \geq 10$ (take logarithms and use $\sum_{2<p\leq x} 1/(p-2) \leq \log\log x + 1.2$), so that $\sum_{d\leq D} g(d)\log d \leq \log D \sum_{d\leq D} g(d) \leq e^{1.2}\log^2 D \leq 0.37\log^2 n$.

If $d > D$: crudely $\#\{\tilde{q} \in W : 2\tilde{q} \equiv b \pmod d\} \leq n/(4d) + 1$. Since $\sum_d g(d)/\sqrt{d} = \prod_{p>2}\big(1 + \frac{1}{\sqrt{p}\,(p-2)}\big) < 2.33$,

$$\sum_{D<d\leq n} g(d)\Big(\frac{n}{4d}+1\Big) \;\leq\; \frac{n}{4}\cdot\frac{2.33}{\sqrt{D}} \;+\; \sum_{d\leq n} g(d) \;\leq\; 0.59\,n^{5/6} \;+\; e^{1.2}\log n.$$

The three secondary terms $0.37\log^2 n$, $0.59\,n^{5/6}$ and $e^{1.2}\log n$, divided by $n/\log n$, are each decreasing for $n \geq 10^8$, and at $n = 10^8$ they total less than $0.51\,n/\log n$; hence $\Sigma \leq 2.92\,n/\log n + 0.51\,n/\log n < 4\,n/\log n$. ∎

Proposition 6.5. *There are absolute constants C^* and N^* (with C^* effective given C_2 of Theorem 2.9 and c_1 of Theorem 2.8, and N^* ineffective) such that: if $n \geq N^*$ and $n \geq C^*L$, then a good prime $\tilde{q} \in W$ exists.*

Neither the statement nor the proof below uses the standing assumption (3.1): the argument depends on (r, s) only through $|s|$, $|rs|$ and the primes dividing rs, so Proposition 6.5 holds as stated for all nonzero integers r, s, of either parity and either sign, including $m(r, s) = 1$ (where the count $\tilde{\omega}$ is 0 and (6.1) is immediate).

Proof. We discard the bad elements of W and count what remains. Set $\tilde{\omega} = \#\{p \leq n : p \mid rs\} \leq \omega(m)$. Assume $n \geq C^*L$ with $C^* \geq 100$ and $n \geq (C^*)^2$, so that $\log(n/C^*) \geq \frac{1}{2}\log n$; by Lemma 2.7,

$$\tilde{\omega} \;\leq\; 1.28\,\frac{L}{\log L} + 1429 \;\leq\; \frac{2.56}{C^*}\cdot\frac{n}{\log n} + 1429. \tag{6.1}$$

(For $L < e$ we have $m \le 15$, so $\omega(m) \le 2$ and (6.1) is trivial; for $e \le L \le \sqrt{n}$, monotonicity of $t/\log t$ on $[e,\infty)$ bounds the first term by $2.56\sqrt{n}/\log n$, which is even smaller. Thus (6.1) holds in all cases for $n \ge (C^*)^2$.)

Bad set 1: $\tilde q \mid s$. At most $\tilde\omega$ elements of W.

Bad set 2: primes $\tilde q$ for which many representations meet rs. For $\tilde q \in W$ put $M = 2\tilde q - 1 \in (n/2,\, n]$, an odd number (the range by the choice of W). Let $\mathrm{Bad}(M)$ be the number of ordered representations $M = q_1 + q_2 + q_3$ in primes with some $q_i \mid rs$. Then

$$\mathrm{Bad}(M) \;\le\; 3 \sum_{\substack{p \mid rs \\ p \le M}} r_2(M-p).$$

The term $p = 2$ contributes r_2 of an odd number, at most 2, so at most 6 per $\tilde q$. For odd p the argument $k = M - p$ is even; when $k < \sqrt{n}$ we use $r_2(k) \le k \le \sqrt{n}$, and when $k \ge \sqrt{n}$ Theorem 2.9 with $k \le n$ and $\log^2 k \ge \frac14 \log^2 n$ gives $r_2(k) \le 4C_2\, S(k)\, n/\log^2 n$. Summing over $\tilde q \in W$ and using Lemma 6.4 for each fixed p (with $b = 1 + p$):

$$\sum_{\tilde q \in W} \mathrm{Bad}(2\tilde q - 1) \;\le\; 6|W| \;+\; 3\,\tilde\omega\left(\tfrac{\sqrt{n}}{2} + 1\right)\sqrt{n} \;+\; 12\, C_2 \frac{n}{\log^2 n} \cdot \tilde\omega \cdot C_3 \frac{n}{\log n}.$$

By Theorem 2.8, for n large every odd $M \in (n/2, n]$ has

$$r_3(M) \;\ge\; c_1 M^2/\log^3 M \;\ge\; \tfrac{c_1}{4}\, n^2/\log^3 n \;=:\; T.$$

Let $E = \{\tilde q \in W:\ \mathrm{Bad}(2\tilde q - 1) \ge \frac12 r_3(2\tilde q - 1)\}$; then $|E| \le 2T^{-1} \sum_{\tilde q} \mathrm{Bad}$, i.e.

$$|E| \;\le\; \frac{8 \log^3 n}{c_1 n^2}\Big[6|W| + 2\tilde\omega n + 12\, C_2 C_3\, \tilde\omega\, \frac{n^2}{\log^3 n}\Big] \;=\; \frac{96\, C_2 C_3}{c_1}\,\tilde\omega \;+\; o\Big(\frac{n}{\log n}\Big),$$

the secondary terms being $O(|W| \log^3 n/n^2) + O(\tilde\omega \log^3 n/n) = O(\log^2 n)$. By (6.1),

$$\frac{96\, C_2 C_3}{c_1}\,\tilde\omega \;\le\; \frac{246\, C_2 C_3}{c_1\, C^*} \cdot \frac{n}{\log n} \;+\; \frac{96 \cdot 1429\, C_2 C_3}{c_1}.$$

Now choose $C^* = \max\{100,\ 4920\, C_2 C_3/c_1\}$, so the first term is at most $0.05\, n/\log n$, and let N^* be large enough that: $n \ge \max\{10^8, (C^*)^2\}$; $2\tilde q - 1 \ge M_0$ for all $\tilde q \in W$; the constant 1429 of (6.1) is at most $0.004\, n/\log n$, so that $\tilde\omega \le 0.03\, n/\log n$; and each of the three remaining terms in the bound for $|E|$ — the constant $96 \cdot 1429\, C_2 C_3/c_1$ and the two secondary terms — is at most $0.01\, n/\log n$. Then, using Lemma 6.3,

$$\#\{\text{good } \tilde q\} \;\ge\; |W| - \tilde\omega - |E| \;\ge\; \Big(0.15 - 0.03 - 0.05 - 0.04\Big)\frac{n}{\log n} \;>\; 0$$

(the 0.03 covering Bad set 1, the 0.05 the main term of $|E|$, and the 0.04 its three remaining terms). Any $\tilde q \in W$ outside both bad sets satisfies (G1), and (G2) because $r_3(2\tilde q - 1) - \mathrm{Bad}(2\tilde q - 1) \ge \frac12 r_3(2\tilde q - 1) > 0$: some representation avoids the primes of rs entirely. $\square$

Remark 6.6. The averaging over $\tilde q$ is not cosmetic. For a single fixed M the worst-case bound $S(M - p) \ll \log\log n$, multiplied by up to $L/\log 2 \asymp n/C$ possible p, exceeds $r_3(M)$ by a factor $\asymp \log n \log\log n/C$ — and even the sharpened count of Lemma 2.7 (attained when rs is a primorial) leaves a factor $\asymp \log\log n/C$: no fixed choice of M can be proved to work by these estimates alone. Averaging restores the loss because, as $\tilde q$ varies over a full window of primes, the singular series $S(2\tilde q - 1 - p)$ is bounded on average (Lemma 6.4); and the sharpened count $\tilde\omega \ll L/\log L$ of Lemma 2.7 — distinct prime divisors cannot all be small — supplies exactly the factor $\log n$ that the trivial bound $\tilde\omega \le L/\log 2$ would lose. As stressed in the introduction, the quantifier order is what makes the averaging legitimate: r and s are given before $\tilde{q}$ is chosen.

7. Proof of the Main Theorem

Proof of the Main Theorem. Let $C = \max\{4.65,\ C^*\}$ and $N_0 = \max\{381700,\ N^*\}$ (the maxima record what the effective cases alone require) with C^*, N^* as in Proposition 6.5, and suppose $n \geq \max\{N_0, C \log m\}$. By the reductions of §3 we may assume (3.1); the targets $z = 1$ and z a transposition are done there. Let $z \in S_n$ be any other target.

If z is even, Proposition 4.1 applies. If z is odd, then $\delta(z)$ is odd; since z is not a transposition, $\delta(z) \geq 3$ (Lemma 2.1(3)). If $\mathrm{m}(z) \geq n/2$, Proposition 5.1 applies. Otherwise $\mathrm{m}(z) < n/2$, and Proposition 6.5 together with Lemma 6.2 applies. In every case $z = x^r y^s$ for some $x, y \in S_n$. □

8. Optimality of the logarithmic shape

Brenner, Evans and Silberger showed that on A_n the constant of their theorem cannot be taken to be 1: for $r = s = n!$ every value of $x^r y^s$ on A_n is trivial, while $\theta(n) = \log m < n$ for infinitely many n [4, proof of Theorem 3′]. Both exponents there are even, so this says nothing about the symmetric-group problem, in which one exponent must stay odd. The following variant respects the parity hypothesis.

Proposition 8.1. *Let $n \geq 3$, let $r = \mathrm{lcm}(1, 2, \ldots, n)$ and let s be the odd part of r. Then r is even, s is odd, $\log m(r, s) = \theta(n)$, and every value of $x^r y^s$ on S_n has all cycle lengths powers of 2. In particular no 3-cycle is a value, and $x^r y^s$ is not universal on S_n.*

Proof. Every cycle length $\ell \leq n$ divides r, so by the power-of-a-cycle fact quoted before Lemma 2.2 every cycle of every $x \in S_n$ collapses to fixed points: $x^r = 1$ identically, and the values of the word are exactly the s-th powers. Write a cycle length of y as $\ell = 2^a b$ with b odd. Then $b \leq n$ gives $b \mid r$, and b odd gives $b \mid s$ (s is the odd part of r), while s odd gives $\gcd(\ell, s) = b$; so, by the same fact, the cycle contributes b cycles of length 2^a to y^s. Thus every cycle length of y^s is a power of 2, and a 3-cycle ($n \geq 3$) is not an s-th power. Finally, the primes dividing rs are exactly the primes up to n, so $\log m(r, s) = \theta(n)$. □

Corollary 8.2. *There is no N_0 such that, for all nonzero integers r, s with at least one odd, the word $x^r y^s$ is universal on S_n whenever $n \geq \max\{N_0, \log m(r, s)\}$. Thus the constant C of the Main Theorem cannot be taken to be 1 (nor, a fortiori, smaller), and the logarithmic shape is optimal for S_n just as for A_n.*

Proof. There are infinitely many integers n with $\theta(n) < n$ (see [16, p. 67], as used in the proof of [4, Theorem 3′]). Given a candidate N_0, pick such an $n \geq \max\{N_0, 3\}$ and take r, s as in Proposition 8.1: then $n \geq \max\{N_0, \log m(r, s)\}$, yet $x^r y^s$ is not universal on S_n. □

9. Concluding remarks

The two 1986 *Proceedings* papers and Merzlyakov's problem set a precise program: a degree bound for $x^r y^s$ logarithmic in $m(r, s)$, uniform in the word, on each of A_n and S_n, together with the matching lower bound. With the Main Theorem and §8, both halves of the symmetric-group case are now closed, forty years after the problem was posed.

Beyond the statement, we would single out the diagnosis of §6.1 as a contribution in its own right: Proposition 6.1 locates the difficulty of Problem 10.32 on the divide between binary and ternary additive prime problems — two cycle lengths force a twin-prime-type constellation, out of reach, while a factor shape with three prime summands moves the problem to ternary Goldbach, where Vinogradov's theorem applies. The analytic devices themselves are classical — the three-primes lower bound, an upper-bound sieve, the averaging of singular series over a window — and we claim no novelty for any of them individually. What may transfer is the arrangement: a three-primes theorem relative to an excluded set of primes that is fixed before the window is chosen, with the adversary defeated on average rather than pointwise, and the count $\omega(m) \ll \log m/\log\log m$ closing the final factor. That pattern — choose the factor shape to fit the arithmetic, then let an exact factorization criterion certify it — may be useful, as may the dyadic windows of §5, in other covering problems in symmetric groups where prescribed prime-avoidance is the obstacle.

Remark 9.1 (Shapes of the factors). The proof gives more than universality: every $z \in S_n$ is expressed with factors of an explicit and very restricted shape — two prime-length cycles for even z; an odd-prime cycle times a $(2^a \cdot \text{prime})$-cycle for odd z of large support; and a type-$(q_1, q_2, q_3, 1)$ permutation times a full even cycle on an N-point subset for odd z of small support. The analysis in §1 explains why some change of shape in the last case is not an artifact of the method: for odd targets with bounded $\delta(z)$ and adversarial r, s, the Herzog–Kaplan–Lev constraints force the two cycle lengths within bounded distance of each other while avoiding prescribed primes — twin-prime territory (Proposition 6.1). It is the passage from two factors' worth of freedom to three summands' worth — from binary to ternary Goldbach — that makes the small-support case provable.

Remark 9.2 (Effectivity). The constant C is an explicit number once the sieve constant C_2 and Vinogradov constant c_1 are fixed from the cited sources; no attempt has been made to optimize it. The threshold N_0 is ineffective solely through M_0 of Theorem 2.8. Replacing Theorem 2.8 by the explicit three-primes estimates underlying Helfgott's proof [8] would, at the cost of numerical work orthogonal to the ideas here, render N_0 effective as well. By contrast, targets that are even or move at least $n/2$ points need only Propositions 4.1 and 5.1: for these, $n \geq \max\{381700,\, 4.65 \log m\}$ suffices, fully explicitly.

Remark 9.3 (Constants). The case constants 4.5 and 4.65 come from windows of relative width $\asymp 1/4$, which is the natural floor of the present placement: sharper Chebyshev bounds (Dusart's, for instance) push both constants toward 4, but not below it, without a structural change. Brenner, Evans and Silberger [4] obtain every $c > 8/5$ for A_n by a finer argument in which whole ranges of primes dividing m are played against each other, and §8 shows $c > 1$ is required for S_n as well. Locating the optimal constant — for the effective cases somewhere in $(1, 4.65]$, and for the theorem as a whole tied to the Vinogradov bookkeeping of §6 — remains open, as does an S_n analogue of the sharp 8/5 threshold of [4].

Remark 9.4 (Verification). Beyond Remark 2.5, two computations were run against this paper's own argument, chiefly to guard against misquotation of the imported criteria and against slips in composition-order conventions: the condition chains fed to Theorems 2.3 and 2.4 in §§4–6 were checked on 600 random instances (n, r, s, z) with $n \leq 3000$, and the small-support construction of Lemma 6.2 was executed end to end in S_{14}–S_{16} on 400 random targets, the identity $x^r y^s = z$ holding exactly in every instance. The lower-bound construction of §8 was likewise verified exhaustively for $n \leq 7$. The scripts, including a rigorous certification of the numerical constants of Lemma 6.4, are available at `https://github.com/AviKndr/universal-words` (`validate.py`).

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